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Paleomagnetism of Cretaceous volcanic rocks from central Chile and implications for the tectonics of the Andes

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ABSTRACT

The mean direction of remanent magnetization in well-dated, structurally and petrologically simple Cretaceous ash-flow tuffs from the central valley of Chile, near Santiago, indicates that the region has undergone a small (about 14°) clockwise rotation. Two other paleomagnetic studies on Cretaceous rocks from coastal Chile give a similar result. We tentatively attribute these rotations to dextral shear caused by oblique subduction of the Nazca plate beneath South America. In contrast, Cretaceous rocks from Peru seem from paleomagnetic evidence to have been rotated counterclockwise relative to the stable interior of the continent. This sense of apparent rotation agrees with the sense of deflection of geologic trends from north-south to north-northwest at about the Peru-Chile border (the Arica deflection). On the basis of these observations, several authors have suggested that the Arica deflection is an orocline. However, uncertainty in all such paleomagnetic interpretations is introduced by what seems to be a very streaked distribution of Cretaceous reference poles from the South American craton. The cause of this streaking is not known.

INTRODUCTION

Models for the tectonic evolution of the North American Cordillera proposed in the past decade (e.g., Jones et al., 1982) lean heavily on paleomagnetic data. Two kinds of data are involved: (1) high-quality reference poles from the craton (e.g., Irving and Irving, 1982; Gordon et al., 1984) and (2) equally high-quality poles from tectonically "suspect" terranes within the orogen (e.g., Champion et al., 1984; Hillhouse and Gromme, 1984). For North America, paleomagnetic information accumulated for roughly 20 years before a pattern emerged that had unmistakable tectonic implications. Once a critical mass of paleomagnetic data was reached, however, the method helped revolutionize Western Cordilleran geotectonics.

We have begun to study paleomagnetism in the South American Cordillera, in the expectation that a tectonic synthesis with paleomagnetic input will eventually emerge. In this paper we report paleomagnetic results for volcanic rocks of Cretaceous age from the central valley of Chile, near Santiago, and show that they can be interpreted in terms of an interesting tectonic model. We also examine several other recent paleomagnetic studies in light of present knowledge of the Mesozoic reference path for South America.

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Figure 1 is a sketch map showing the geomorphic provinces and geography of central Chile. The main features shown are the Coast Range, reaching elevations of about 1000 m, and the main range of the Andes, which in this region contains several peaks over 5000 m. Lying between the two mountain ranges is the central valley of Chile, a topographic low mostly underlain by recent alluvium, but containing low hills composed of mid-Cretaceous and younger volcanic rocks. This configuration of geomorphic features extends for over 1000 km to the south of the area shown in Figure 1.

The distribution of Cretaceous and early Tertiary rocks in the central valley, near Santiago, is sketched in Figure 1, after the *Mapa Geologico de Chile.* Also shown are sampling localities for the K/Ar age determinations reported in Table 1. The rocks sampled for paleomagnetic and K-Ar studies are mostly welded ash-flow tuffs of felsic composition that form sheets a few metres to a few tens of metres thick. These sheets probably covered large regions when they were first erupted; erosion has given them a patchy current distribution that makes correlation difficult. These ash-flow sheets have been gently tilted (to a maximum of about 30°) but are not highly deformed. Neither are they much altered; where sampled (mostly in quar-

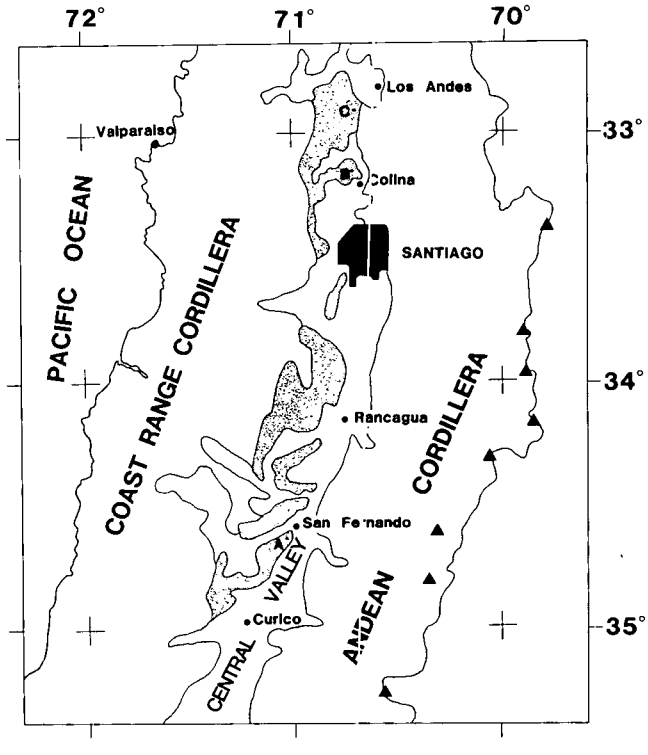


Figure 1. Location map showing some features of geology and geography of central Chile. Sampling areas: A—San Fernando ash-flow tuffs; B, C—Chacabuco ash-flow tuffs; see Table 1. Stipple pattern shows area of outcrop of Late Cretaceous–early Tertiary volcanic rocks, after *Mapa Geologico de Chile.*

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TABLE 1. K-Ar ANALYTICAL DATA, CRETACEOUS ASHFLOW TUFFS, CENTRAL CHILE

Sample No.	Material	Sample weight, g	K %	RA	PRA	m.y. ± 2 s.d.	Location
921-3	Biotite	0.6075	5.892	101.70	46	97 ± 3	A (34.62S, 70.99W)
C-82-70	Biotite	0.2747	6.973	13.48	78	108 ± 2	A (34.62S, 70.99W)
C-82-70	Plagioclase	2.9085	0.606	11.20	85	104 ± 5	A (34.62S, 70.99W)
C-82-10	Plagioclase	5.6404	0.777	8.86	96	64.6 ± .9	B (33.17S, 70.67W)
C-82-11	Plagioclase	4.2783	0.728	9.10	90	70.7 ± 1.4	B (33.17S, 70.67W)
C-82-12	Plagioclase	0.7674	0.856	8.97	56	59.5 ± 3.6	B (33.17S, 70.67W)
C 82 71	Plagioclase	2.3463	0.731	9.37	78	72.5 ± 4.0	B (33.17S, 70.67W)
RD-803	Plagioclase	5.4432	0.492	5.77	90	66.4 ± .4	C (32.90S, 70.70W)
RD-801	Plagioclase	2.9912	1.023	11.66	78	64.5 ± 1.4	C (32.90S, 70.72W)
HP-235	Plagioclase	2.2643	1.060	12.70	91	67.8 ± 1.3	C (32.95S, 70.70W)

NOTE: ^{40}K decay constants $\lambda_e + \lambda_{e'} = 0.581 \times 10^{-11}$/yr; $\lambda\beta = 4.962 \times 10^{-11}$/yr; $^{40}K/K = 1.167 \times 10^{-4}$. RA = radiogenic argon, x 10^{-11} mol/g. PRA = percent radiogenic argon. A,B,C are sampling localities shown in Figure 3.

ries) they appear fresh and reassuringly uncomplicated. The paleomagnetic results show that they are excellent recorders of the ancient geomagnetic field. As indicated in Table 1, there are two ages of ash-flow tuffs: about 100 Ma near San Fernando and about 65 Ma near Chacabuco, approximately 180 km to the north.

PALEOMAGNETIC RESULTS

We collected from 5 to 8 oriented samples at each of 40 sites. Samples were drilled and oriented in situ; we used both sun and magnetic compass for orientation. Natural remanent magnetization (NRM) was measured on a spinner magnetometer, and using a 2-axis reciprocating tumbler, we subjected all samples to alternating field demagnetization. In general, the rocks responded to laboratory treatment in a straightforward and uncomplicated way; except for a weak present-field overprint, none showed evidence of multiple magnetizations. Thermal cleaning also was applied to representative samples, but it failed to uncover any serious complications in the magnetic record. Site 84CK07 is a very fine-grained red sediment associated with the ash-flow tuffs. Because of the presence of hematite, it was cleaned entirely by thermal methods.

Our paleomagnetic results are listed in Tables 2 and 3 and plotted in Figure 2. Mean directions from the San Fernando area (Fig. 2, top) are well grouped and all of normal magnetic polarity, whereas those from the Chacabuco area (Fig. 2, bottom) are more scattered and include a few lower-hemisphere directions. These differences between the two data sets probably reflect differences in the ancient geomagnetic field; radiometric ages of the San Fernando rocks place them within the mid-Cretaceous long normal interval, whereas ages of the Chacabuco rocks fall during a period of frequent transitions in geomagnetic polarity (Harland et al., 1982). We tentatively interpret Chacabuco directions CK2, 4, and 12 as representative of transitional states of the geomagnetic field, or as "excursions," and we exclude them from all calculations of mean directions or poles. Mean poles and directions from the San Fernando and Chacabuco areas are significantly different at the 95% confidence level.

TABLE 2. MEAN DIRECTIONS OF MAGNETIZATION FOR ASH-FLOW TUFFS FROM THE SAN FERNANDO AREA, CHILE

Site	N	Dec	Inc	α_{95}	k	Demag Level
84CK13	6	10.1°	-43.2°	3.8°	400	40
14	7	8.6	-36.5	2.6	513	10
15	7	3.9	-40.1	6.2	96	20
16	8	0.5	-43.0	4.8	135	40
20	8	15.3	-47.0	3.6	232	30
21	8	351.3	-53.5	2.5	494	25
22	8	23.5	-59.2	5.8	91	25
23	8	15.5	-53.4	2.5	482	40
24	7	22.2	-55.4	3.4	311	40
25	8	22.8	-47.1	3.7	231	100
26	8	15.9	-43.6	2.1	697	60
27	8	33.0	-31.8	3.0	351	30
28	8	18.1	-37.4	3.0	342	80
29	8	358.3	-57.1	5.8	92	20
30	8	6.4	-57.2	6.7	70	15
31	8	11.8	-76.1	4.5	155	25
32	8	26.9	-47.3	2.8	407	40
33	8	359.1	-58.6	8.6	42	20
83SF01	7	49.1	-43.2	1.6	1417	30
02	7	54.1	-36.5	3.4	313	40
03	6	340.0	-57.8	7.2	87	40
04	7	36.7	-50.2	8.2	56	60
Mean	22 16.8	-50.2	6.0	27.7		
Ex Dir		2.7	55.8	5.0		
Rotation			14.1 9.0°		Flattening	-5.6 ± 6.2°

Note: All directions corrected for dip. N = number of samples; Dec = declination; Inc = inclination; α_{95} = radius of circle of 95% confidence about the mean direction, assuming the Fisher (1953) distribution; k = best estimate of the precision parameter of Fisher (1953); Demag level = peak value of alternating field used to magnetically clean the site, in mT. Ex Dir = the expected direction of the sampling area, assuming a dipole magnetic field with pole at 87.3°S, 234.6°E, (α_{95} = 5.0°). Rotation = the difference between observed and expected direction, with 95% confidence limits (see Beck, 1980; Demarest, 1983). Flattening = the difference between expected and observed inclination.

The mean of 22 sites from the San Fernando area almost certainly provides adequate sampling of the ancient geomagnetic field, but the same may not be true of the 15 sites sampled near Chacabuco. The exposures sampled near San Fernando are parts of a single large topographic feature, within which the various ash-flow tuff layers are arranged in a simple monoclinal sequence. Thus, barring undetected faults, we can be sure that we have sampled 22 separate cooling units. The shape of the distribution of San Fernando site mean directions also suggests that geomagnetic secular variation has been adequately sampled. The case is different for the Chacabuco data, however. In the Chacabuco area the exposures are discontinuous and scattered over a distance of several tens of kilometres; therefore, we cannot be sure that we have not sampled the same unit (and thus the same geomagnetic field direction) more than once. In fact, the tendency for site-mean directions to "cluster" (shown in Fig. 2, bottom) suggests that we may have done that. We

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can demonstrate that we have sampled at least five cooling units, and we have sampled across at least one polarity transition. Nevertheless, the amount of time sampled at Chacabuco is not certain; therefore, the degree to which secular variation has been averaged is not clear. Certainly five independent measurements are inadequate. For this reason the Chacabuco mean may have very little tectonic significance, and we do not discuss it further. We hope to do more field work in the Chacabuco area in the near future.

INTERPRETATION

To interpret our paleomagnetic results in terms of possible tectonic displacements we must have an apparent polar wandering (APW) path for stable South America. Much work has been done on the South American Cretaceous, but as shown below, the results are somewhat perplexing.

Valencio et al. (1983), Schult and Guerreiro (1979), and other authors showed that as much as 30° of APW relative to South America took place during the Cretaceous. This makes choice of a reference pole difficult. For comparison with our San Fernando results, we calculated a 100 Ma South American reference pole from 20 virtual geomagnetic poles (VGP) for the Sierra del los Condores Group, Argentina (Vilas, 1976); the La Yesera Formation, Argentina (Valencio et al., 1977); and the volcanic rocks from Cabo de Santo Agostinho, Brazil (Schult and Guerreiro, 1980). From evidence cited in the originals, these rocks appear to have ages between about 85 and 115 Ma. For the pole calculation we used only VGP based on mean directions for three or more oriented samples per site, and we excluded any VGP for which the circle of 95% confidence about the mean direction exceeded 15°. Location, confidence statistics, and scatter statistics for this reference pole are given in Table 2.

From this reference pole we next calculated the expected direction, rotation, and flattening values given at the bottom of Table 2. The San Fernando rocks show a clockwise rotation of about 14°, which is significant at the 95% confidence level. We tentatively attribute this rotation to dextral shear caused by right-oblique convergence between South America and the Nazca plate (Pilger, 1983). However, the mean inclination of the San Fernando rocks is somewhat shallower than expected (F = –5.5°), a fact that suggests transport from the north. This is the wrong direction for transport by dextral shear. This flattening value is not significant at the 95% level.

OTHER ANDEAN PALEOMAGNETIC RESULTS

Palmer et al. (1980) studied the paleomagnetism of ash-flow tuffs near La Serena, in northern Chile. The rocks they sampled are certainly Cretaceous, but evidence of low-grade metamorphism and K-Ar dates that range from 59 to 92 Ma indicate that some thermal or chemical remagnetization may have taken place. Thus, the exact time of magnetization of the La Serena volcanics is not known. If the La Serena rocks acquired their magnetization in the same magnetic field as the San Fernando rocks (their similar lithologies and geologic settings suggest that this is likely), then they also have undergone a small but statistically significant clockwise rotation (7.7 ±6.2°).

Turner et al. (1984) also studied Cretaceous rocks in northern Chile. Despite encountering complex magnetizations, they isolated a characteristic remanent magnetization from sandstone interbeds in the dominantly conglomeratic Coloso Formation. The age of this magnetization, however, appears to be in question. The Coloso Formation itself is described as Early Cretaceous, but its magnetization is at least partly the result of postdepositional processes and could be younger. Figure 3 shows that the Coloso Formation pole agrees reasonably well with poles for other Cretaceous units from Chile, in that it lies to the right of the present south pole as viewed from the sampling area. Thus, the Coloso Formation also may show clockwise rotation.

Cretaceous rocks from coastal Peru, on the other hand, all seem to have undergone counterclockwise rotation. There are two recent studies. Heki et al. (1984) measured remanent magnetization in Cretaceous volcanic and sedimentary rocks distributed along the coast of Peru between lat 6° and 15°S. Again, the age of the magnetization of some of these rocks is poorly known. Heki et al. calculated a reference pole from five studies on Argentine and Brazilian Cretaceous rocks with which to compare their results, and concluded that their sampling areas record counterclockwise rotations ranging from about 19° to 52°. A similar result was obtained by May and Butler (1985), who studied remagnetized sedimentary and volcanic rocks from the Early Cretaceous Puente Piedra Formation, near Lima. Remagnetization probably occurred during intrusion of the nearby Coastal Batholith of Peru, at about 90 Ma. May and Butler compared their direction with an expected direction calculated from the 90 Ma pole of Irving and Irving (1982). They concluded that the Lima area had rotated about 20° counterclockwise since the mid-Cretaceous. Representative Peruvian poles are shown in Figure 3.

TABLE 3. MEAN DIRECTIONS OF MAGNETIZATION FOR ASH-FLOW TUFFS FROM THE CHACABUCO AREA, CHILE

Site	N	Dec	Inc	α	k	Demag level
84CK01	8	5.7	-45.9	3.0	350	20
02*	7	197.1	6.5	11.9	28	30
03	7	341.4	-68.1	4.0	228	20
04*	8	99.0	-30.5	8.7	41	20
05	7	341.4	-64.5	5.7	112	20
06	8	207.3	83.8	2.5	496	20
07	5	269.6	79.9	6.3	115	350 deg
08	8	335.5	-67.8	4.3	129	30
09	8	310.2	-43.5	5.4	108	10
10	8	355.7	-51.4	6 6	70	30
11	7	7.9	-60.8	6.1	64	30
12*	8	0.7	3.5	7.4	56	40
83LT01	8	347.2	-46.9	4.0	189	20
02	7	350.2	-60.7	2.3	879	40
03	7	351.2	-62.1	2.8	830	60
04	7	351.5	-42.4	3.2	346	60
05	8	346.3	-71.2	2.4	524	25
06	7	345.0	-41.4	5.0	318	50
Mean	15	348.9	-60.9	8.3	22.3	

Note: See Table 2 for explanation of abbreviations. Site 84CK07 was cleaned by thermal technique.
* Site not used to calculate formation mean.

ROLE OF THE REFERENCE CURVE

From Figure 3 it seems certain that there is a real difference between Cretaceous poles from Chile and those from Peru. If the reference pole for the South American craton calculated for this paper is correct, and if it is of the appropriate age to serve as reference for all the Andean studies shown in Figure 3 (a doubtful assumption), then portions of coastal Chile have been rotated clockwise by small angles, and portions of coastal Peru have experienced somewhat larger counterclockwise rotations.

Two indirect (and not particularly compelling) lines of evidence argue that the small Chilean rotations may be the result of dextral shear. (1) Long range-paralleling faults have been mapped in northern Chile (St. Amand and Allen, 1960) and in the Chilean Lake district (Herve, 1976). Available evidence sug-

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gests, but apparently does not prove, that these are fossil right-lateral strike-slip faults that were active at least into the Miocene. (2) Convergence between the Nazca plate and South America was right-oblique during the Late Cretaceous and early Tertiary (Pilger, 1983). This is in the correct sense to cause dextral shear in the South American continental margin.

On the other hand, the change in tectonic, geologic, and topographic trends seen in the Andes near the Peru/Chile border (the Arica deflection) suggests that counterclockwise rotations in Peru are due to a post–middle Cretaceous rigid rotation of the entire Peruvian Andes. That is, the paleomagnetic evidence and the changes in "trends" taken together imply that the Arica deflection is an orocline, in the sense of Carey (1955). Exactly what might have driven oroclinal bending of the Andes is not obvious (May and Butler, 1985, discussed the process further).

From the foregoing it appears that paleomagnetism has already made a straightforward and dramatic contribution to the tectonics of the Andes. However, things may not be quite so simple. Figure 3 also shows a series of small numbered dots, which represent Cretaceous paleomagnetic poles from Argentina and Brazil; the poles provide the data from which the APW path for stable South America is constructed. They are numbered in approximate chronological order, from oldest (1) to youngest (14). Some of these poles are from the eastern foothills of the Andes, so they might conceivably be influenced by local tectonics. Many, however, are from parts of South America that have been completely stable since the Cretaceous. As with any collection of paleomagnetic poles, some of these poles are more reliable than others. In particular, the age of magnetization of many of the rocks represented here is not well determined; therefore, little importance should be placed on the chronological sequence indicated by the numbers in Figure 3.

However, what seems to be certain is that the APW path for "stable" South America swung through an arc of several tens of degrees one or more times during the Cretaceous. The pattern looks very much like a small-circle distribution about an Euler pole located near the southern tip of South America. By contrast, Cretaceous poles from the Andes, which are even more conspicuously streaked, form a distinctively different pattern. We are at a loss to explain these distributions of poles.

Whatever its cause, the tectonic consequences of this streaked pattern of reference poles are important. Unless the age of magnetization of an Andean rock unit is very well established, it is impossible to know with which part of the reference APW path to compare its paleomagnetic direction. For instance, to interpret Peruvian poles CV, PP, and DA, May and Butler (1985) and Heki et al. (1983, 1984) used reference poles that fall within a few degrees of the present spin axis. Compared to such poles the Peruvian studies are highly discordant and have a large counterclockwise rotation. However, the actual age of magnetization of many of the Peruvian rocks is not well known. If poles CA, PP, and DA ac-

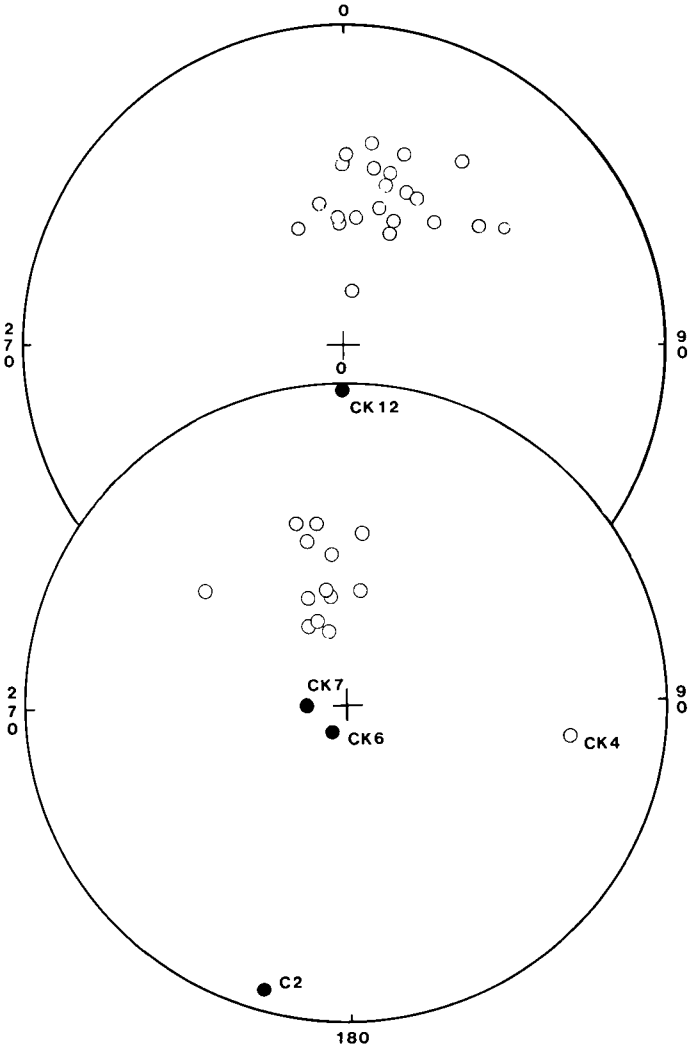


Figure 2. Equal-area projection of site-mean directions for ash-flow tuffs near San Fernando (top) and Cuesta de Chacabuco (bottom). Open symbols indicate upper hemisphere; solid symbols indicate lower hemisphere.

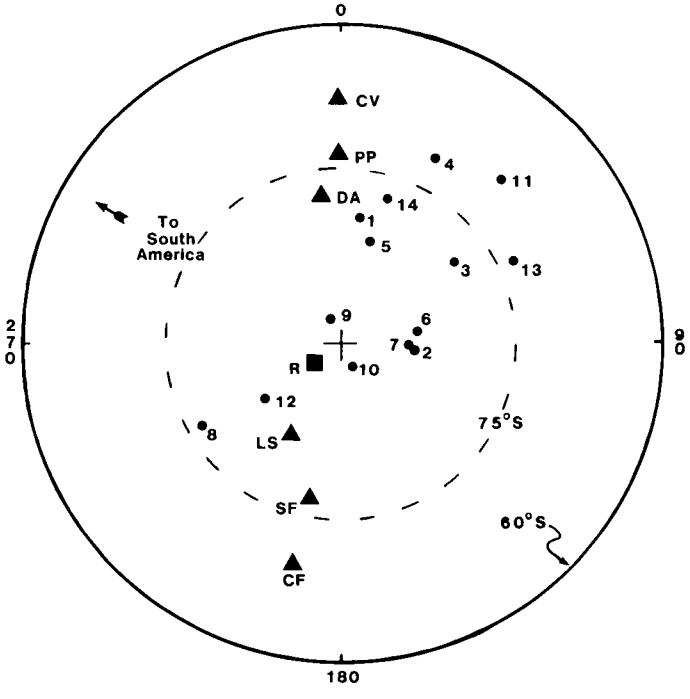


Figure 3. Southern hemisphere, equal-area plot of selected paleomagnetic poles for Cretaceous rocks from South America. Square is reference pole given in Table 2 and discussed in text. Triangles = paleomagnetic poles from Andes: CV, DA—Heki et al. (1983); PP—May and Butler (1985); LS—Palmer et al. (1980); SF—this paper; CF—Turner et al. (1984). Dots = paleomagnetic poles from eastern foothills of Andes, or from craton: 1—Linares and Valencio (1975); 2—Bellieni et al. (1983); 3—Creer (1962); 4—Mendia (1978); 5—Valencio (1972); 6—Schult and Guerreiro (1979); 7—Vilas (1976); 8—Valencio et al. (1977); 9—Schult and Guerreiro (1980); 10—Valencio et al. (1977); 11—Valencio et al. (1983); 12—Opdyke and MacDonald (1973); 13—Vilas and Valencio (1978); 14—Linares and Valencio (1975). Note expanded scale.

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tually lie in the time interval 140–115 Ma, then they probably should be compared with a reference pole constructed from craton poles 1–5. Relative to this pole the Peruvian studies are much less discordant, and the case for the Arica orocline is far less compelling. For instance, with composite pole 1–5 as reference, study PP yields a counterclockwise rotation of only 7.2°, but it has a mean inclination that is more than 12° too steep. Thus, if the magnetization of the Puente Piedra Formation happens to be significantly older than concluded by May and Butler (1985), a case could be made for some 800 km of post–middle Cretaceous northward transport of coastal Peru, relative to stable South America, accompanied by an insignificant amount of counterclockwise rotation. We think this scenario is highly unlikely, but it illustrates the importance of the reference APW path for tectonic interpretation.

CONCLUSIONS

The mean direction of mid-Cretaceous ash-flow tuffs from the San Fernando area of the Chilean central valley suggests that the area has experienced a clockwise rotation of roughly 14°, relative to the stable interior of South America. The time of rotation, the size of the rotated block, and the precise mechanism of rotation are unknown. Post–middle Cretaceous clockwise rotations are also found in rocks from coastal and central Chile as far north as the Peruvian border. However, north of Arica, where tectonic and geographic trends change from north-south to northwest-southeast, paleomagnetic measurements indicate that the rocks have rotated counterclockwise, some through angles exceeding 45°. Again, the precise timing, scale, and mechanism of rotation are unknown. These interpretations are complicated by the curious streaked distribution of Cretaceous reference poles for the South American craton; therefore, it is difficult to calculate expected paleomagnetic directions for Cretaceous rocks from the orogen with much confidence. This distribution of Cretaceous poles for the craton probably has important tectonic (or geomagnetic) implications; however, it too does not seem to be well understood. Quite clearly, our principal conclusion must be that a great deal of work remains to be done.

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